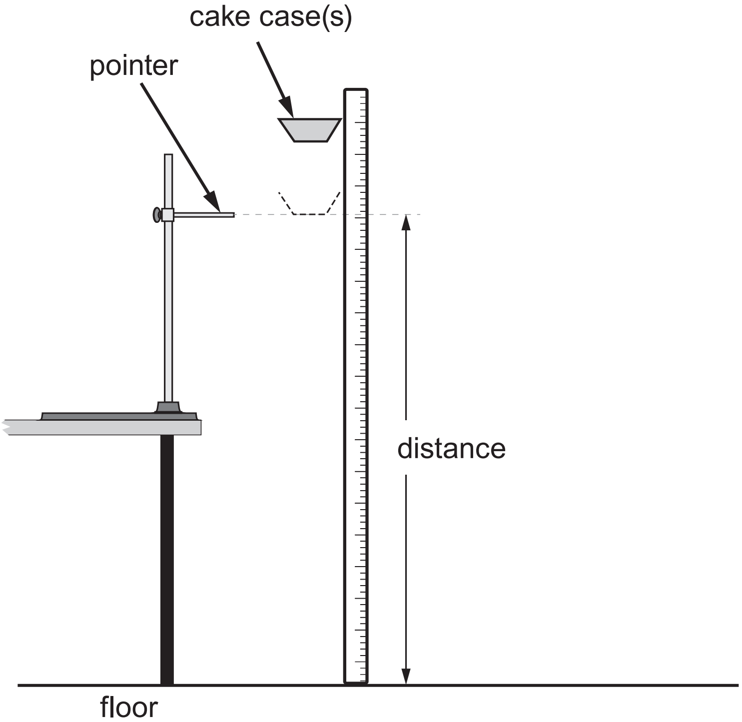
Examiner only

2. A group of students investigate if the terminal speed of falling paper cake cases depends on their mass. They follow the method given below.



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1. Set up a pointer in the clamp stand and set it 1.50 m above the ground.
2. Take a single cake case and record its mass using a balance.
3. Drop the cake case from 20 cm above the pointer.
4. Use a stopwatch to record the time it takes to fall from the level of the pointer to the floor.
5. Repeat steps 3 and 4 another four times.
6. Repeat steps 3-5 with extra cake cases in a stack.

(a) (i) Give **one** reason why the students let the cake case fall for 20 cm before starting the stopwatch. [1]

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(ii) Give **two** reasons why the students take more than one repeat reading. [2]

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(b) The following data are collected. The students assume that each of the cake cases has a mass of 0.5 g.

Number of cake cases	Mass of cake cases (g)	Mean time taken for cake cases to fall 1.50 m (s)	Terminal speed (m/s)
0	0		0
1	0.5	0.94	1.60
2	1.0	0.67	2.24
3	1.5	0.59	2.54
5	2.5	....	2.88
6	3.0	0.51	2.94

(i) **Complete the table.** Use an equation from page 2 to calculate the missing value of the mean time. *Space for workings.* [2]

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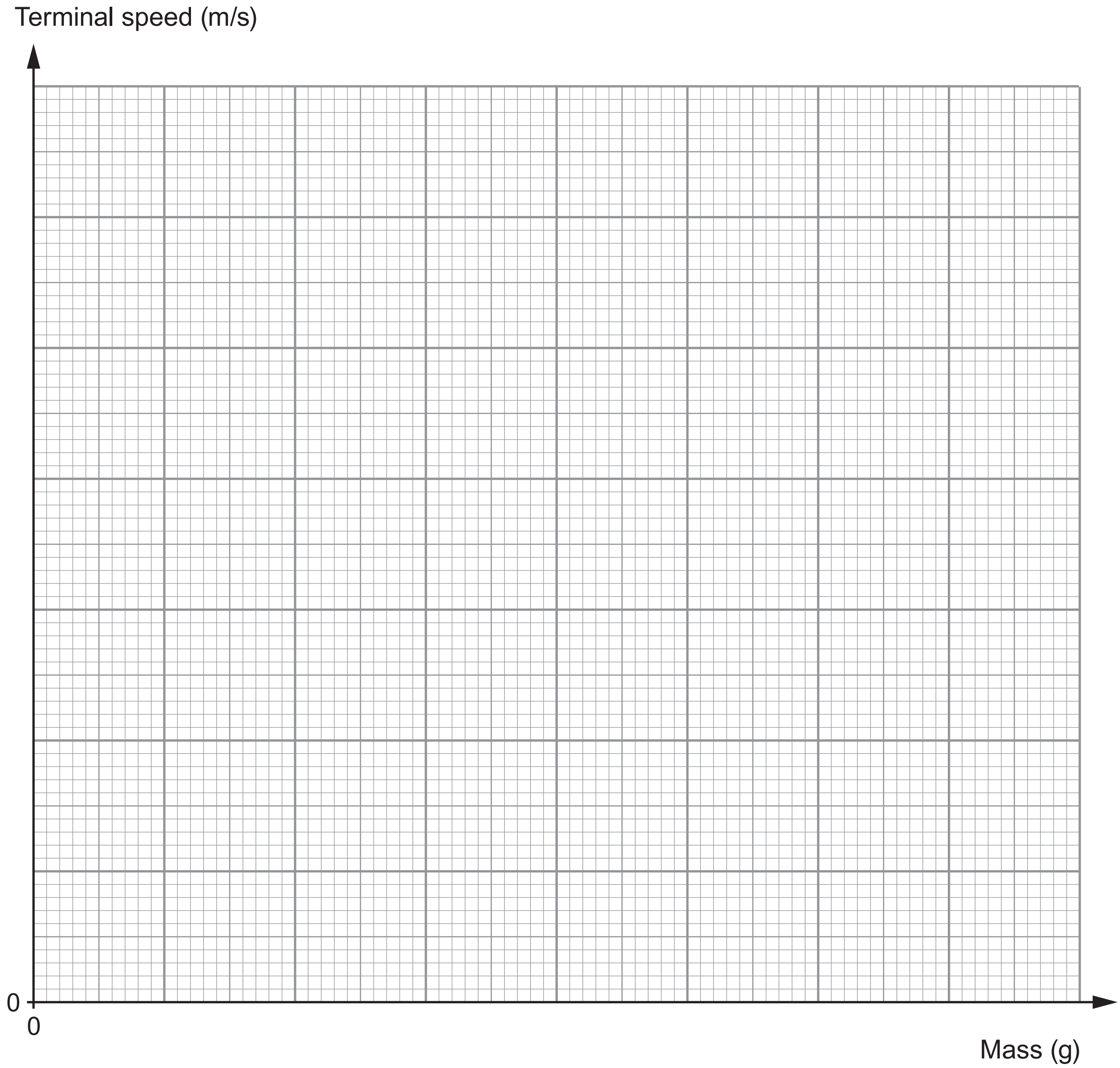
(ii) Plot the data on the grid below and draw a suitable line. [4]



(iii) One of the students suggests that the terminal speed will always increase by a factor of 1.4 if the mass is doubled. The student finds that when the mass doubles from 0.5 g to 1.0 g this suggestion is true. Explain if this is true for the other masses. [3]

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(c) Apart from taking more repeat readings, suggest one way in which the method could be improved to collect better quality data **and** explain how the improvement would give better data. [2]

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(d) Explaining your reasoning and using an equation from page 2, calculate the size of the air resistance force acting on a stack of 5 cake cases when travelling at terminal speed. (Gravitational field strength, g = 10 N/kg). *Space for workings.* [3]

Air resistance = ... N

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